Name: Balasathya . S
Roll. No: EE24B093

# Digital Systems - Tutorial 1

$0.8\times2 = 1+0.6$
$0.6\times2 = 1+0.2$
$0.2\times2 = 0+0.4$
$0.4\times2 = 0+0.8$
$0.8\times2 = 1+0.6$

$(11100.11001)_2$ 11100

1. Convert $(28.8)_{10}$ to binary. Show the result up to five places after the binary point.

7 2. Determine the positive value of $x$ if $(211)_x = (152)_8$. $2\times8^0+5\times8^1+1\times8^2 = 1\times x^0+1\times x+2\times x^2 \Rightarrow x=7$

$(9.75)_{10}$ 3. Convert $(1001.110)_2$ to Decimal. Assume input is unsigned. $1\times2^3+1\times2^0+1\times\frac{1}{2}+1\times\frac{1}{4} = 9.75$

$(0.4162)_8$ 4. Convert $(0.528)_{10}$ to Octal.

$(11000.10)_2$ or $(-7.5)_{10}$ 5. Calculate the sum of 1010.011 and 1110.001. Assume the numbers are in 2's complement format.

$(108A.738)_{16}$ 6. Convert $(7562.45)_{10}$ to hexadecimal.

$(0010000)_2$ or $(16)_{10}$

7. Let $X = 1010100$ and $Y = 1000011$. Calculate $X-Y$ and $Y-X$. Assume the input numbers are in 2's complement format.

$(1101111)_2$ or $(-17)_{10}$

$(416825)_{10}$ 8. Convert $(640.7)_8$ to Decimal. $7\times\frac{1}{8}+0\times8^0+4\times8+6\times8^2$

$(0011101)_2$ or $29_{10}$

9. Perform $(+42)+(-13)$ and $(-42)-(-13)$ using binary arithmetic. Select the appropriate number of bits and represent the input numbers in two's complement format.

$(1000011)_2$ or $29_{10}$

$(100001.1)_2$ 10. Represent $(-30.5)$ in Two's Complement form. $-32+1.5 = 100000+1+0.1 = 100001.1$

4. Convert $(0.528)_{10}$ to Octal.

$0.528\times8 = 4+0.224$
$0.224\times8 = 1+0.792$
$0.792\times8 = 6+0.336$
$0.336\times8 = 2+0.688$

6. Convert $(7562.45)_{10}$ to hexadecimal.

| | |
|---|---|
| | 7562 |
| 10 | 472 |
| 8 | 29 |
| 13 | 1 |
| 1 | 0 |

108A.733

$0.45\times16 = 7+0.2$
$0.2\times16 = 3+0.2$

5. Calculate the sum of 1010.011 and 1110.001. Assume the numbers are in 2's complement format.

$1010.011 \rightarrow -8+2+1/4+1/8 = -5.625$
$1110.001 \rightarrow -8+4+2+1/8 = -1.875$
$11000.100 \rightarrow -8+1/2 = -7.5$

-44 -61

7. Let $X = 1010100$ and $Y = 1000011$. Calculate $X-Y$ and $Y-X$. Assume the input numbers are in 2's complement format.

$1010100 + 0111101 = 10010001 \rightarrow X-Y \rightarrow 17$

$1000011 + 0101100 = 1101111 \rightarrow Y-X \rightarrow -17$

9. Perform $(+42)+(-13)$ and $(-42)-(-13)$ using binary arithmetic. Select the appropriate number of bits and represent the input numbers in two's complement format.

$42 \rightarrow 0101010 \Rightarrow -42 \rightarrow 1010110$
$13 \rightarrow 01101 \Rightarrow -13 \rightarrow 10011$

$001010 + 010011 = 011101 \rightarrow 29_{10}$

$1010110 + 0001101 = 1100011 \rightarrow -29_{10}$